

CASE STUDY

ROADS & HIGHWAYS

Detailed Topographical Survey for NH 154A Chamba (Himachal Pradesh)

OVERVIEW

The project was carried out through a detailed topographical survey using drone-based photogrammetry. The data acquisition was undertaken to support feasibility studies and detailed engineering for the rehabilitation and upgradation of the highway. The primary objective was to ensure that high-accuracy terrain and elevation datasets were generated to facilitate design planning, alignment optimization, and infrastructure development in challenging mountainous terrain.

OBJECTIVE

- Capture high-resolution overlapping aerial imagery.
- Establish accurate GCPs.
- Generate high-quality DTM, DSM, orthophoto, and contour datasets.



APPROACH & TECHNOLOGY

APPROACH

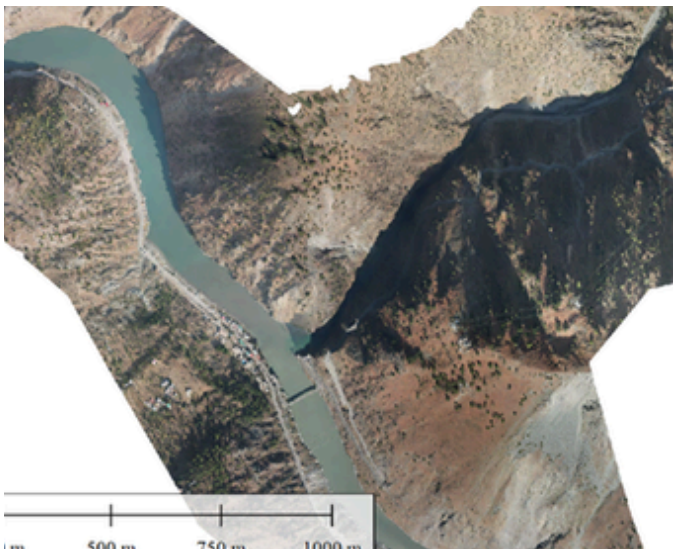
- Overcome the challenges of difficult terrain and limited accessibility.
- Meeting stringent accuracy requirements.
- Systematic and technology-driven approach with utilizing drone-based photogrammetry.

TECHNOLOGY

- Multi-rotor UAV equipped with high-resolution RGB sensors.
- DGPS/RTK methods to achieve sub-meter horizontal and vertical accuracy.
- Advanced photogrammetry software to generate orthophotos, DTM, DSM and contours.

CONCLUSION

The processed data was used to provide critical topographical inputs for tunnel alignment design, approach road profiling, and infrastructure planning. Overall, the methodology was proven to be a cost-effective, time-saving, and accurate solution, and high-quality datasets were delivered.



CASE STUDY

ROADS & HIGHWAYS

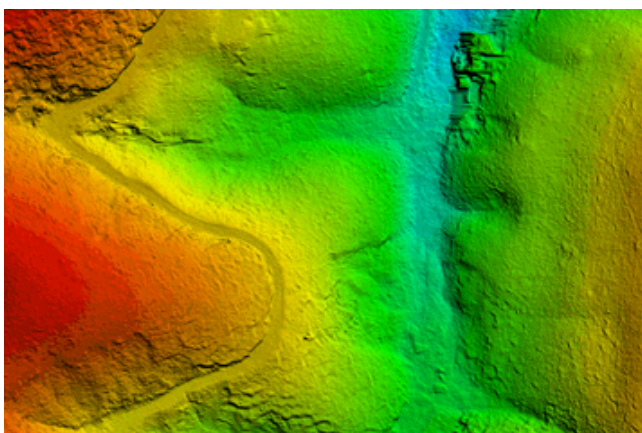
50 km Proposed Alignment in Aizawl, Mizoram

OVERVIEW

This project was carried out through a high-resolution Drone LiDAR Survey for the proposed road alignment in Aizawl, Mizoram, covering approximately 50 km with a minimum 200-meter ROW (100 m on each side of the centerline). The survey was conducted using UAV-based LiDAR technology to ensure that precise terrain and surface information was captured, which was required for design feasibility and engineering analysis.

OBJECTIVE

- Obtain precise LiDAR-based elevation and surface details.
- Generate an accurate DTM, and contour datasets.
- Create orthorectified imagery and classified LiDAR point clouds.



APPROACH & TECHNOLOGY

APPROACH

- The proposed alignment and ROW requirements were reviewed.
- UAV flight paths were planned to ensure complete coverage of the 50 km stretch.
- Raw LiDAR data, GNSS/IMU data, and aerial imagery were collected simultaneously.

TECHNOLOGY

- UAV equipped for long linear corridor mapping.
- High-precision UAV-mounted LiDAR system.
- Dual-frequency GNSS receiver integrated with an Inertial Measurement Unit (IMU).
- Point cloud calibration, Noise filtering, Classification, DTM, orthophoto and contour creation.

CONCLUSION

The Drone LiDAR Survey conducted for the proposed alignment in Aizawl, Mizoram was successful in delivering high-accuracy geospatial data essential for engineering design and feasibility assessment. Advanced UAV-mounted LiDAR, GNSS/IMU navigation systems, and state-of-the-art processing software were utilized, to meet the required engineering standards.



CASE STUDY

RAILWAYS

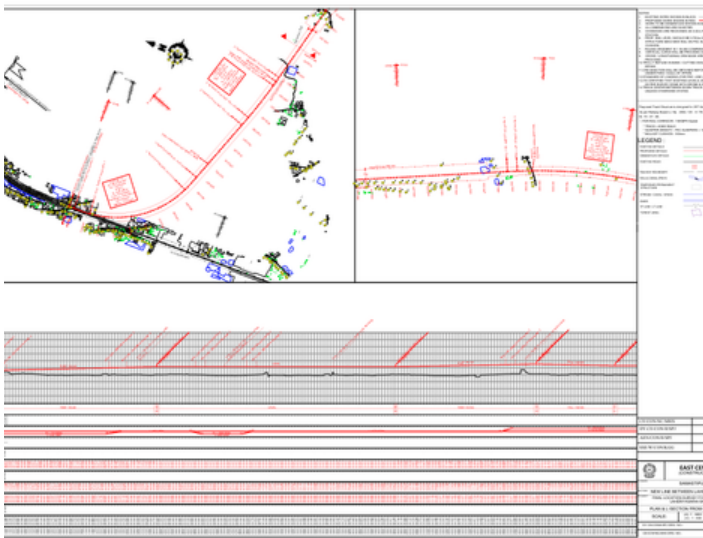
New Railway Line Between Laheriasarai-Saharsa (100 km)

OVERVIEW

A new 100 km railway line connecting Laheriasarai and Saharsa was proposed to provide a shorter, faster, and more efficient transport corridor in North Bihar. At present, 150–180 km is traveled by passengers using indirect routes, resulting in long journey times. With the new line, travel distance and time will be significantly reduced, and regional connectivity across Darbhanga, Madhubani, Supaul, and Saharsa districts will be improved.

OBJECTIVE

- To provide the shortest and most efficient rail connectivity between Laheriasarai and Saharsa.
- To reduce travel distance by 50–80 km.
- To support passenger movement and freight transport.
- To boost regional economic growth.



APPROACH & TECHNOLOGY

APPROACH

- Detailed traffic, economic, and technical feasibility analysis was conducted.
- Alternative route options were evaluated and compared, and the shortest, safest, and most economical alignment was selected.

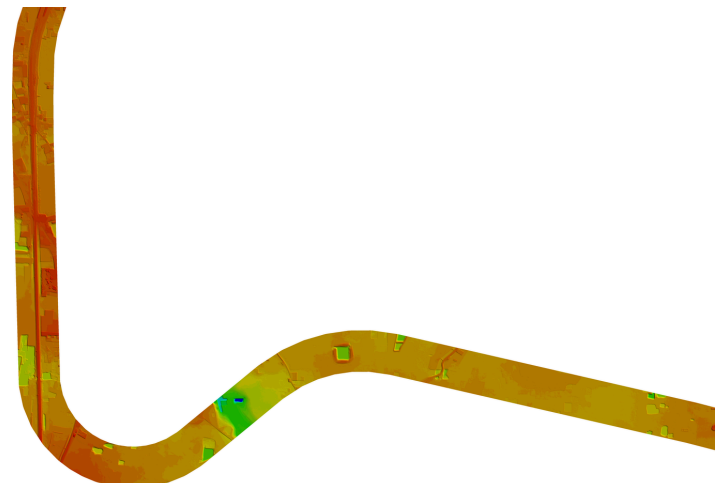
TECHNOLOGY

- Advanced surveying using DGPS, drone mapping, and digital tools like AutoCAD Civil 3D and QGIS for accurate alignment design and terrain modeling.

CONCLUSION

The proposed Laheriasarai-Saharsa new railway line is technically feasible, economically viable, and socially beneficial.

The project assist in providing improved mobility, enhanced freight movement, and overall socio-economic upliftment of the region.



CASE STUDY

RAILWAYS

Detailed Project Report Preparation for Ayodhya Bypass New Rail Line



OVERVIEW

This project was carried out through a high-precision aerial topographical survey for the proposed new railway line between the Salarpur-Masodha section, which is designed to be a bypass for the Ayodhya Cantt Railway Station under North Eastern Railway. The survey is aimed at ensuring that accurate terrain, alignment, and structural data are generated to support detailed engineering design, planning, and construction of the new rail corridor.

OBJECTIVE

- Capture accurate topographical and terrain information.
- Generate detailed base maps, orthomosaics, DTM/DSM, and contour.
- Support the planning of bridges, RUBs, yard layouts, and hydrological assessments.
- Enable precise alignment design, longitudinal profiles, and cross-sections.

APPROACH & TECHNOLOGY

APPROACH

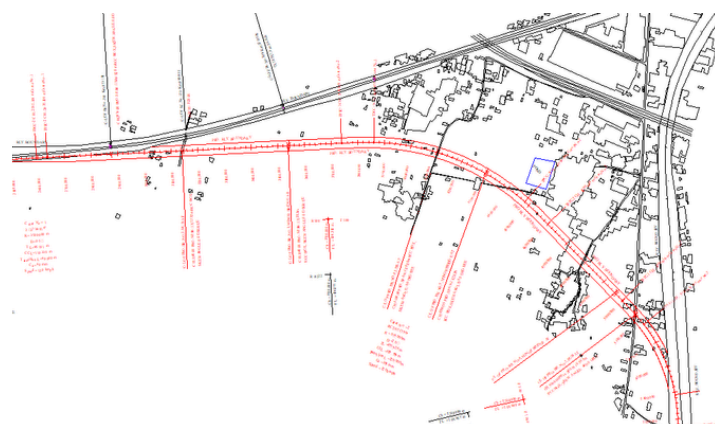
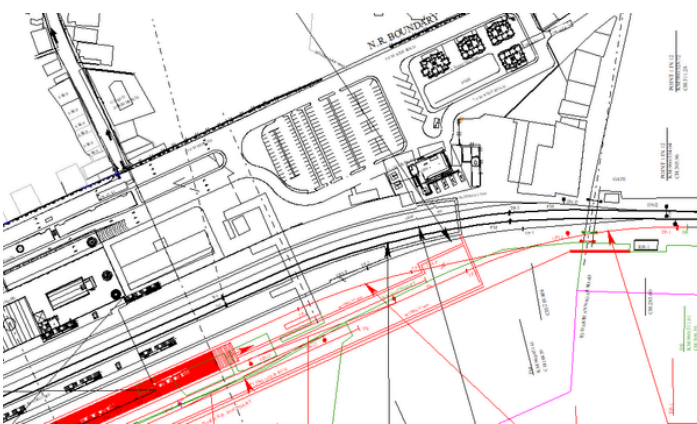
- Survey planning.
- Aerial data acquisition.
- Ground control survey.
- Data processing.
- Engineering drawings and output.

TECHNOLOGY

- UAV equipped with high-resolution RGB sensors.
- Dual-frequency RTK/DGPS receivers for establishing GCPs.
- Advanced processing suites for Dense point clouds, Orthomosaic imagery, DTM, DSM and 10cm contours.

CONCLUSION

The aerial topographical survey for the Salarpur-Masodha bypass was completed, delivering a precise geospatial foundation for the design and development of the new railway corridor. Through the use of UAV-based photogrammetry, GNSS-controlled survey techniques, and robust GIS/CAD engineering tools, high-quality maps, elevation models, alignment designs, and structural planning inputs were successfully generated.



CASE STUDY

RAILWAYS

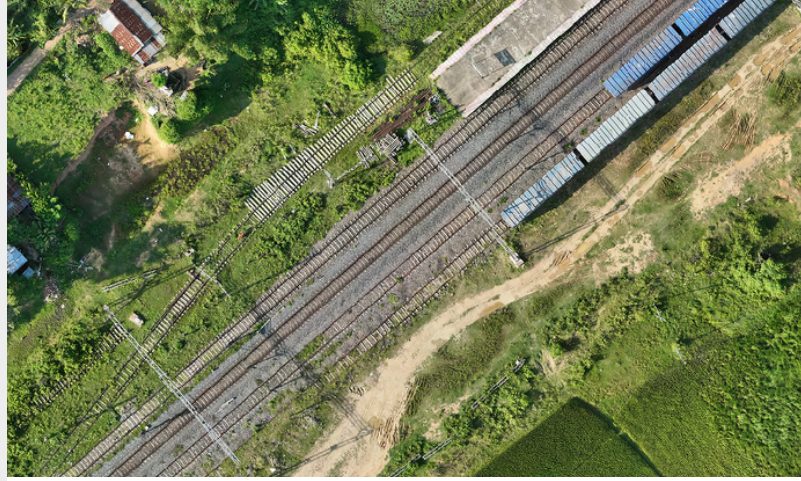
Detailed Survey & Design for Dhansiri-Naojan Doubling of Track (EPC Project)

OVERVIEW

This project was executed through a detailed aerial and ground-based topographical survey to support the EPC (Engineering, Procurement & Construction) work for doubling the railway track between Dhansiri and Naojan stations in Nagaland under Northeast Frontier Railway (NEFR). Accurate terrain, alignment, and hydrological information was provided by the survey, which was essential for engineering design, planning, and construction of the upgraded double-line section.

OBJECTIVE

- Capture precise topographical, leveling, and terrain data through aerial photogrammetry and total station survey.
- Generate DEM, 10cm contours, and engineering base maps essential for track doubling design.
- Support preparation of detailed alignment plans, longitudinal sections, cross-sections, yard/ESP plans, and hydrological assessments.



APPROACH & TECHNOLOGY

APPROACH

- Ground Survey Operations.
- Aerial Data Acquisition.
- Survey Control Network.
- Data Processing & Feature Extraction.
- Engineering Outputs (alignment plan, longitudinal section, cross-sections, yard/ESP plans).

TECHNOLOGY

- Total Station & Digital Leveling.
- GNSS / RTK Survey Integration.
- Photogrammetric Processing.
- Alignment design, cross-sections, feature extraction, and ESP/yard planning.
- Watershed modeling techniques for drainage and hydraulic calculations.

CONCLUSION

The Dhansiri-Naojan EPC track doubling project was supported by a comprehensive aerial and ground survey through which high-accuracy terrain, alignment, and hydrological datasets were delivered. By integrating UAV photogrammetry with precise total station and leveling techniques, reliable topographical inputs for detailed engineering design were ensured.



CASE STUDY RAILWAYS

Final Location Survey for Prayagraj-Naini Doubling of Track

OVERVIEW

This project was being carried out through a high-accuracy Final Location Survey (FLS) for the doubling of the railway track between Prayagraj and Naini stations. Using drone-based photogrammetry and detailed topographical mapping, precise terrain, alignment, and structural information was provided, as required for engineering design and planning. The deliverables are used to support the preparation of alignment plans, ESP/yard layouts, longitudinal sections, and cross-sections for the track-doubling project near Naini Bridge.

OBJECTIVE

- Capture accurate topographical and terrain data using drone-based aerial survey.
- Generate high-resolution orthomosaic, DSM/DTM, and 10cm contour data for engineering-level accuracy.
- To support the preparation of alignment plan, L-section, cross-sections, and ESP/yard plans for track doubling.



APPROACH & TECHNOLOGY

APPROACH

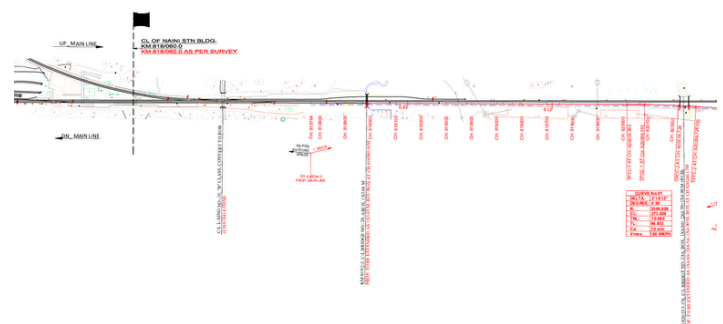
- Survey Planning.
- Aerial Data Acquisition.
- Ground Control Survey.
- Photogrammetric Data Processing.
- Engineering Design Inputs.

TECHNOLOGY

- High-resolution drone system.
- Dual-frequency GNSS receivers for precise GCP marking.
- Dense point cloud generation using photogrammetry.
- Georeferenced orthomosaics.
- Generating DSM, DTM and 10cm contours using photogrammetry.

CONCLUSION

The drone-based Final Location Survey for the track-doubling project was completed, and precise topographical and alignment datasets were delivered, which were essential for final engineering design. Using advanced UAV photogrammetry, GNSS-controlled surveying, and robust CAD/GIS tools, accurate mapping of terrain, corridor features, and existing railway infrastructure was ensured.



CASE STUDY

URBAN DEVELOPMENT

UAV-Based Topographical Survey & Master Planning for SEZ (Mumbai)

OVERVIEW

This project is focused on providing technical services for the preparation of a 30-year Detailed Master Plan and the architectural design for model buildings within the SEZ region. Using drone-based aerial mapping, detailed topographical datasets, terrain models, cross-sections, and feature extractions are produced to support urban planning, infrastructure layout, and architectural concept design.

OBJECTIVE

- Generate accurate, large-scale topographical and terrain datasets using UAV-based photogrammetry.
- Prepare a 30-year Master Plan supported by detailed mapping of buildings, roads, water bodies, utilities, and civic structures.
- Deliver high-quality architectural and planning inputs for model building design within



APPROACH & TECHNOLOGY

APPROACH

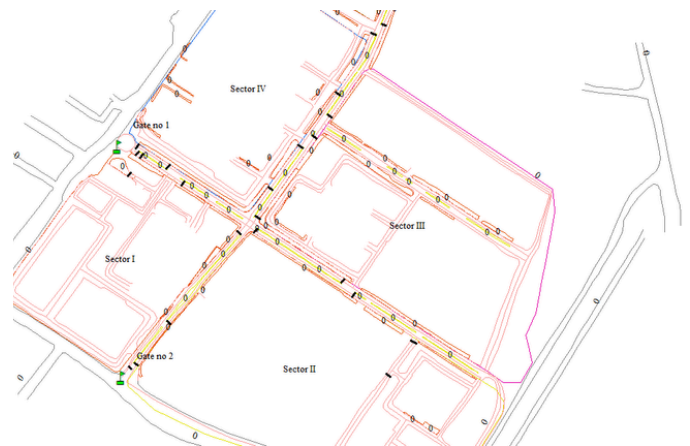
- Pre-Survey Planning.
- UAV Aerial Mapping.
- Ground Control & Field Activities.
- Data Processing & Extraction.
- Master Plan & Architectural Support.

TECHNOLOGY

- UAV Photogrammetry.
- Dual-frequency DGPS/RTK units for control point establishment.
- Photogrammetric Processing (cloud points, DTM, DSM, 3D visualization).

CONCLUSION

A comprehensive spatial foundation for a 30-year Master Plan and detailed architectural design was provided by the UAV-driven topographical survey and technical services for urban development. The creation of accurate maps, 3D models, terrain datasets, and sectional analyses is ensured through the integration of high-resolution aerial mapping, precise DGPS control, and advanced GIS/CAD workflows.



CASE STUDY

SATELLITE BASED SURVEY

Stereo Satellite Mapping for Delhi Tunnel Corridor (NH-148AE)

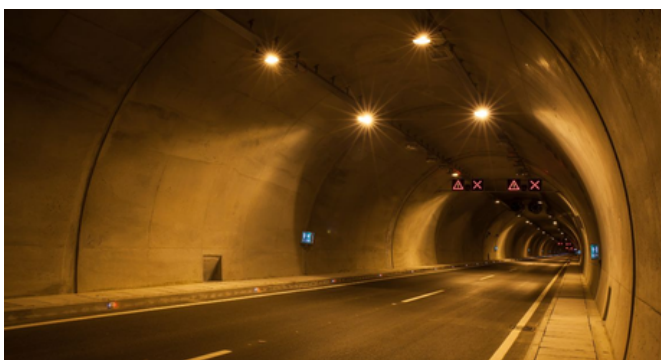


OVERVIEW

High-resolution 50 cm stereo satellite imagery for a 100 sq. km corridor in Delhi was supplied for this project to support the preparation of the Detailed Project Report (DPR) for the proposed road tunnel under NH-148AE. Essential three-dimensional terrain information was provided by the imagery, which was required for planning tunnel alignment, approach roads, and connections to NH-248BB and internal Delhi roads.

OBJECTIVE

- Acquire and supply high-resolution (50 cm GSD) stereo satellite imagery covering the tunnel corridor.
- Generate precise DTM, contours, and ortho imagery to support DPR preparation.
- Deliver accurate topographical and geomorphologic datasets required for preliminary tunnel alignment and feasibility studies.



APPROACH & TECHNOLOGY

APPROACH

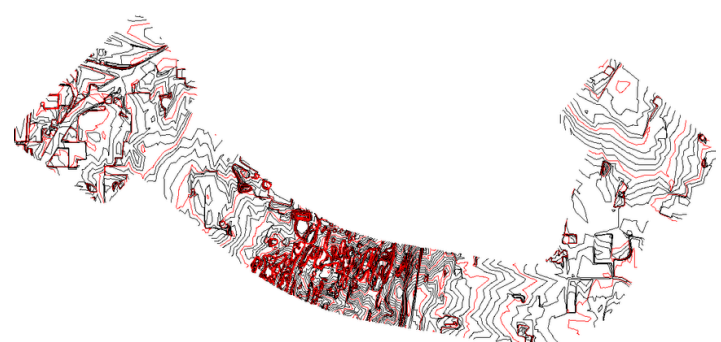
- Data Acquisition (Satellite imagery).
- Establishing Ground Control Points (GCPs) using DGPS.
- Stereo Image Processing.
- Topographic Feature Extraction.
- Quality Assessment & Delivery.

TECHNOLOGY

- High-Resolution Stereo Satellite imagery.
- Establishing precise GCPs for sub-meter geolocation accuracy.
- Photogrammetry & DEM Extraction.
- GIS Mapping (2D base maps, topographic layers, and design-compatible outputs).

CONCLUSION

High-resolution stereo satellite imagery and advanced terrain datasets, which are crucial for planning and designing the tunnel corridor, are delivered by this satellite-based tunnel project. Through precise stereo capture, accurate ground control, and rigorous photogrammetric processing, DTM, contours, and ortho imagery were produced, forming the foundation for detailed project reporting and pre-construction engineering.



CASE STUDY

SOLAR POWER

3D Reality Model For Solar planning of the Paper Mill

OVERVIEW

A 3D model of the Paper Mill area was created to provide a detailed representation of the terrain, infrastructure, and built-up environment for solar planning. High-resolution aerial imagery and topographical data were utilized to ensure accurate modeling for planning, design, and operational analysis. The project was aimed at supporting engineering, environmental assessment, and urban development activities in the region.

OBJECTIVE

The primary objective of the project was to generate a precise 3D model of the Paper Mill area that could be used for planning, design validation, and simulation of proposed infrastructure developments. Accurate spatial and structural information was intended to be provided to aid decision-making and reduce planning risks.



APPROACH & TECHNOLOGY

The project was executed using UAV-based photogrammetry and LiDAR surveys to capture high-resolution imagery and elevation data. Precise GNSS/RTK control was applied to ensure spatial accuracy. Advanced GIS and 3D modeling software were used to process the data, generate terrain models, build 3D structures, and produce orthomosaics, DEMs, and contours. Digital Elevation Models (DEMs), Digital Surface Models (DSMs), orthomosaics, and contours were produced to represent the area in detail. The 3D model was developed through integration of these datasets into 3D modeling platforms, where existing infrastructure and terrain features were accurately mapped.

CONCLUSION

A high-accuracy 3D model of the Paper Mill area was successfully produced, providing reliable terrain and infrastructure datasets. The deliverables were used to support detailed engineering, planning, and operational analysis, ensuring informed decision-making and improved project efficiency.

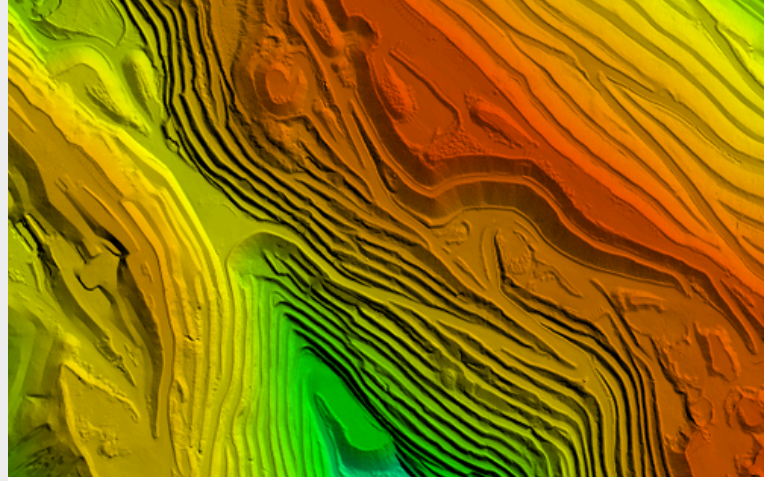


CASE STUDY MINE

UHD Reality modeling and risk analysis of A Narayan Mine

OVERVIEW

A detailed UHD reality model of the A Narayan Mine was developed to provide accurate spatial and structural representation of the mining area. High-resolution terrain and surface data were captured to support mine planning, monitoring, and risk assessment activities. The project was aimed at enabling efficient volume estimation, crack identification, and hazard evaluation.



APPROACH & TECHNOLOGY

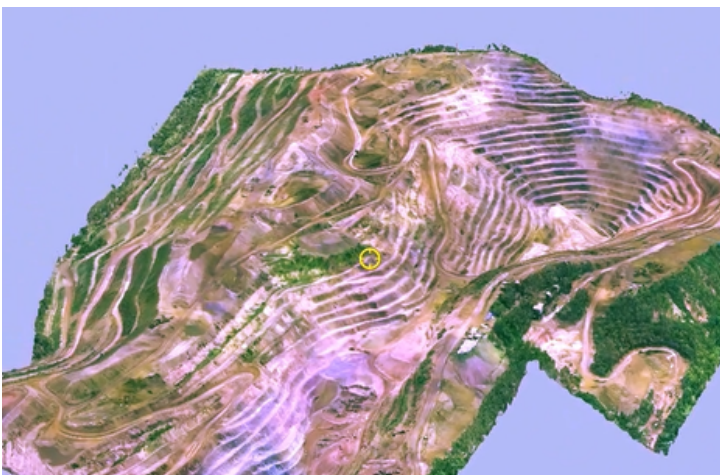
UAV-based photogrammetry and LiDAR surveys were employed to acquire high-resolution aerial imagery and elevation data. Ground control points were established using GNSS/RTK surveying to ensure spatial accuracy. Collected data was processed to generate DSM, DTM, orthomosaics, and point clouds. Crack identification and risk zones were mapped through advanced image analysis and GIS workflows. Volume estimation was performed using 3D modeling software, and all outputs were validated for accuracy and reliability.

OBJECTIVE

The primary objective was to generate precise DSM, DTM, and orthomosaic datasets, along with a 3D model, to support volume estimation, crack detection, and risk analysis for the Mine. Accurate geospatial and structural information was intended to be provided for safe and efficient mining operations.

CONCLUSION

A high-accuracy 3D model of the A Narayan Mine was successfully produced, along with DSM, DTM, orthomosaics, and analytical outputs for crack detection, risk assessment, and volume estimation. The deliverables were utilized to support informed decision-making, safe mining practices, and efficient resource management.



CASE STUDY INTERNATIONAL

UAV based Planimetric Mapping for Dual Sites in the Kingdom of Saudi Arabia

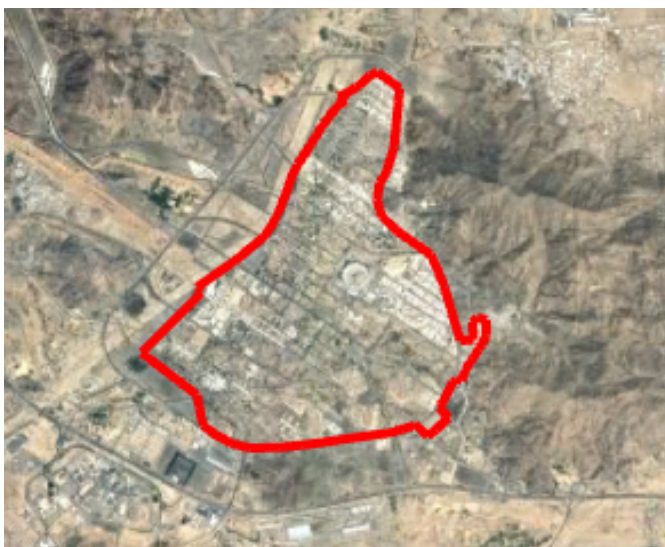


OVERVIEW

High-precision digitization and feature extraction of two designated areas in the Kingdom of Saudi Arabia (KSA), covering a total area of 21.25 sq.km, was carried out in this project. An accurate and up-to-date geospatial database was generated by utilizing drone-based aerial imagery with 5 cm Ground Sampling Distance (GSD).

OBJECTIVE

- A high-accuracy and comprehensive geospatial database.
- Digitization and detailed feature extraction of 5 cm GSD UAV imagery.
- All physical, infrastructural, and environmental elements — with special focus on dense urban zones.



APPROACH & TECHNOLOGY

APPROACH

- A systematic photogrammetric and GIS workflow.
- Ensure spatial accuracy, completeness, and consistency of extracted features across both project areas.
- Quality-controlled digitization and multi-stage validation.

TECHNOLOGY

- Advanced UAV data processing.
- GIS software was used to create feature layers for all identified elements, including buildings and built structures, transportation networks, utility and street infrastructure.

CONCLUSION

The created geo-database was made compatible with municipal asset management systems and visualization overlays on orthophotos for validation and planning. Smart city initiatives, urban planning, and infrastructure development were supported by the project.



CASE STUDY INTERNATIONAL

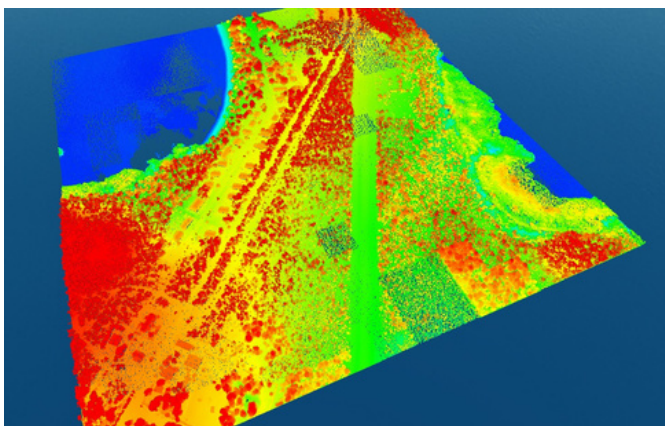
LiDAR data Classification and Terrain Modeling for a Region in Australia

OVERVIEW

The processing and analysis of LiDAR data was carried out to generate high-accuracy terrain products for a selected area in Australia. The LiDAR point cloud was classified into meaningful land-surface and object categories, and key elevation-based datasets, including the Digital Terrain Model (DTM), Digital Surface Model (DSM), and contour layers, were developed. Advanced LiDAR classification techniques were applied to separate ground, vegetation, buildings, and other surface features, ensuring that a clean and reliable dataset was produced.

OBJECTIVE

- LiDAR point cloud data classification.
- Generating high-quality elevation products.
- Ground and non-ground features separation.
- Precise, consistent, and industry-standard datasets intended to enhance the understanding and interpretation of the region's terrain characteristics.



APPROACH & TECHNOLOGY

APPROACH

- Automated classification and manual editing.
- Excessive volumes of sensor noise were excluded.
- Permanent or semi-permanent residential, commercial, and industrial buildings were included in the buildings class.

TECHNOLOGY

- Points with spuriously high or low Z values were classified as noise.
- TerraScan was used for classification and noise filtering.
- Photogrammetry based softwares used for DTM/DSM generation, contour creation, and validation.

CONCLUSION

A detailed understanding of both surface features and bare-earth terrain was provided by the final outputs, enabling improved decision-making for planning, design, land assessment, and resource management. Rigorous quality checks were applied to ensure that all deliverables met industry standards and aligned with the accuracy requirements of the project.

